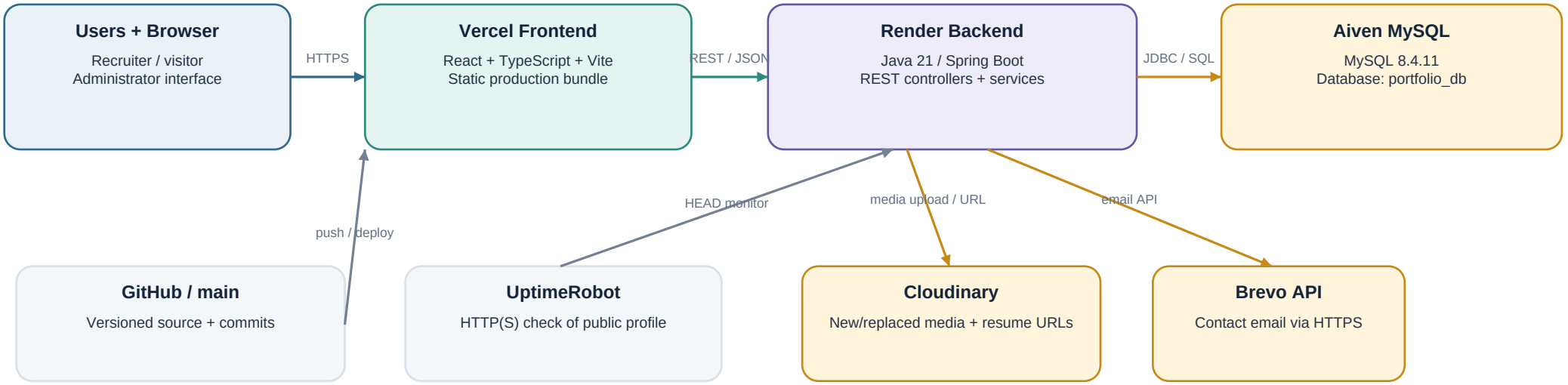
Portfolio Website | Production Architecture

Executive view of the deployed system, runtime calls, data services and delivery pipeline



REQUEST PATH

Browser loads the Vercel bundle, then React/Axios calls the Render API. Spring Security applies the route rules before application logic accesses Aiven MySQL. Media uploads and contact notification email leave the backend through HTTPS integrations.

DEPLOYMENT PATH

A source push to GitHub has historically been followed by provider deployments. Confirm the actual connected branch, Root Directory, Build Command and Start Command in each dashboard before relying on automation.

CURRENT RELIABILITY NOTE

Render Free can sleep; Aiven Free can power off for inactivity. Backups and alerting matter, and free tiers do not guarantee zero downtime.

EVIDENCE LIMIT

Source tree/dashboard were not available to this audit. Unverified details are marked in the editable guide and evidence checklist.

Technology Reference

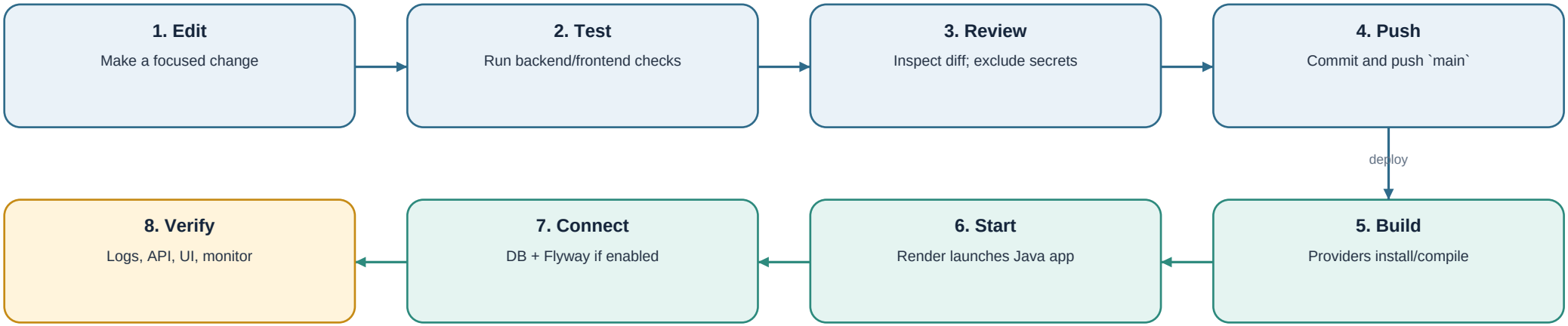
Project-specific purpose, alternatives and confidence; not a generic technology glossary

Technology / tool	Category	Where used	Why used in this project	Alternatives / verification
React + TypeScript	Frontend	Public UI + admin UI	Reusable components and typed UI contracts.	Vue / Angular. Confirm current package versions.
Vite + npm	Frontend build	`frontend` build	Fast dev/build tool; creates static production assets.	Webpack / Next.js. Verify Vercel output (`dist`).
Axios	API client	Frontend API requests	Shared request defaults, longer GET timeout and one transient retry.	Fetch API. Current helper/config reported implemented.
Java 21 + Spring Boot	Backend runtime/framework	Render REST API	Controllers, services, security and persistence integration.	Node.js / .NET. Version reported; verify `pom.xml`.
Spring Security + JWT	AuthN / AuthZ	Public-route allowlist; admin APIs	Public routes are explicit; remaining requests require authentication.	Session auth / OAuth2. Inspect token claims and role checks.
BCrypt + stateless sessions	Security	Authentication provider	Password hashing and stateless REST authentication.	Argon2 / server session. BCrypt + STATELESS in supplied config.
Maven + Flyway	Build / migrations	Backend tests and DB schema	Reproducible test workflow; versioned migration history.	Gradle / Liquibase. Tests passed; V1-V7 reported.
MySQL 8.4.11 + Aiven	Database / hosting	`portfolio_db`	Managed relational persistence outside backend process.	PostgreSQL. Exact tables and FKs need source audit.
Vercel + Render	Cloud hosting	Frontend / backend	Managed static delivery plus Java web runtime.	Netlify / Railway / other hosts. Check project settings.
Cloudinary	Media storage	New/replaced images + resume	Keeps current uploads outside ephemeral Render disk.	S3-compatible storage. Older assets were not bulk-migrated.
Brevo HTTPS API	Email integration	Contact notification	Uses outbound HTTPS rather than SMTP.	Other transactional email APIs. Delivery tested in project history.
GitHub + UptimeRobot	Version control / monitoring	`main` + public API monitor	Commit history and availability alerts.	GitLab / other monitor. Confirm current monitor status.

Not documented as active components: Docker, Kubernetes, Terraform, AWS, or a project-owned CI workflow. Add them only if the repository or hosting settings prove they are used.

How the Application Builds, Starts and Deploys

Separate local validation, provider deployment, runtime startup and healthy service checks



CONFIRMED LOCAL COMMANDS

Frontend production build

```
npm --prefix .\frontend run build
```

Backend test suite

```
mvn -f .\backend\pom.xml test
```

DASHBOARD VALUES TO CONFIRM

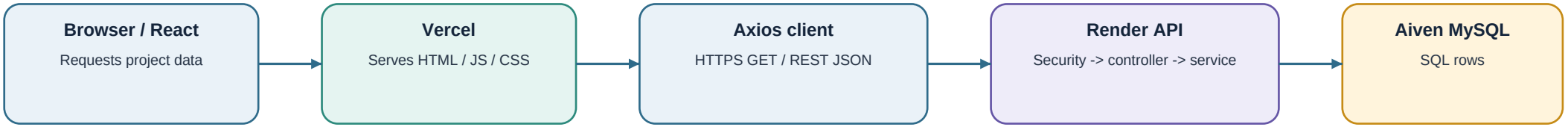
Vercel: Root Directory (`frontend` reported), build command, output directory (`dist` expected for Vite), production API base URL and production branch. Render: Build Command, Start Command, Java version, health-check path, instance plan, latest deployed commit, runtime environment variables. Do not substitute guesses for actual dashboard values.

Flyway has a V7 media cleanup queue migration in the project history. Startup migration behavior depends on the actual dependencies/configuration and logs. Never manually edit migration history just to make startup pass.

Runtime Request Flows

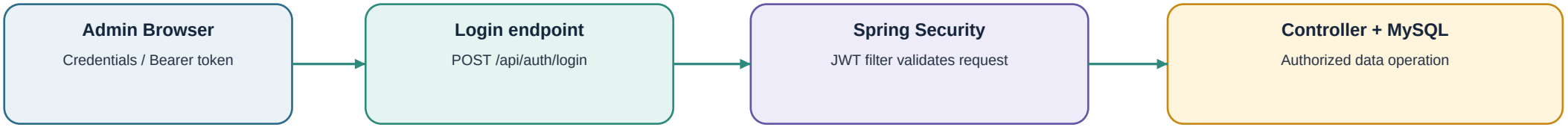
The browser never accesses MySQL directly; the backend owns authorization, database access and third-party credentials

A. Public Projects request



Return path: MySQL rows -> Spring serializes JSON -> HTTP 200 response -> Axios parses response -> React renders project cards. A failed database connection often becomes a backend 5xx response or prevents Spring Boot from finishing startup.

B. Administrator login and protected CRUD

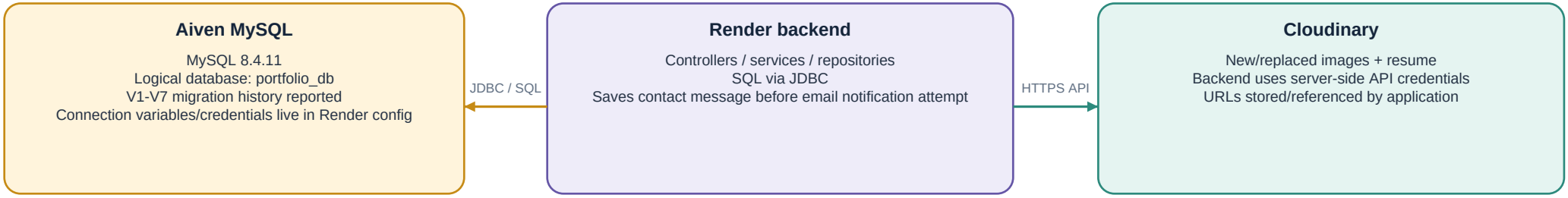


The supplied config permits POST /api/auth/login and explicitly permits selected public GET endpoints; all other requests fall through to `.anyRequest().authenticated()`. The JWT issuance contract, token expiry/claims, and any role-specific checks must be confirmed from the auth/controller code.

Observed monitoring change: commit 379c312 added public HEAD access for /api/public/profile and added HEAD to the CORS method list. The user-reported live HEAD request returned HTTP 200; confirm UptimeRobot records the monitor as Up.

Data Persistence and Media Storage

Distinguish durable database records, managed media URLs and the legacy local-file migration gap



Logical concept	Known role	Schema confidence
Profile / Site Info / Social Links	Public identity, metadata and links	Resource names observed; exact table names/columns require migrations/entities.
Projects / Skills / Education / Experience	Portfolio content	Resources observed from public endpoints/admin interface history.
Certifications / Achievements	Credentials and milestones	Admin/public resources observed; exact FKs unknown.
Contact Messages	Stores visitor messages before notification attempt	Save-before-email sequence confirmed from project history; table schema not inspected.
Admin identity	Authenticates admin access	BCrypt auth provider configured; table/role mapping not inspected.
Media Cleanup Queue	Tracks candidate media cleanup work	V7__media_cleanup_queue.sql reported; query/retention behavior needs source review.

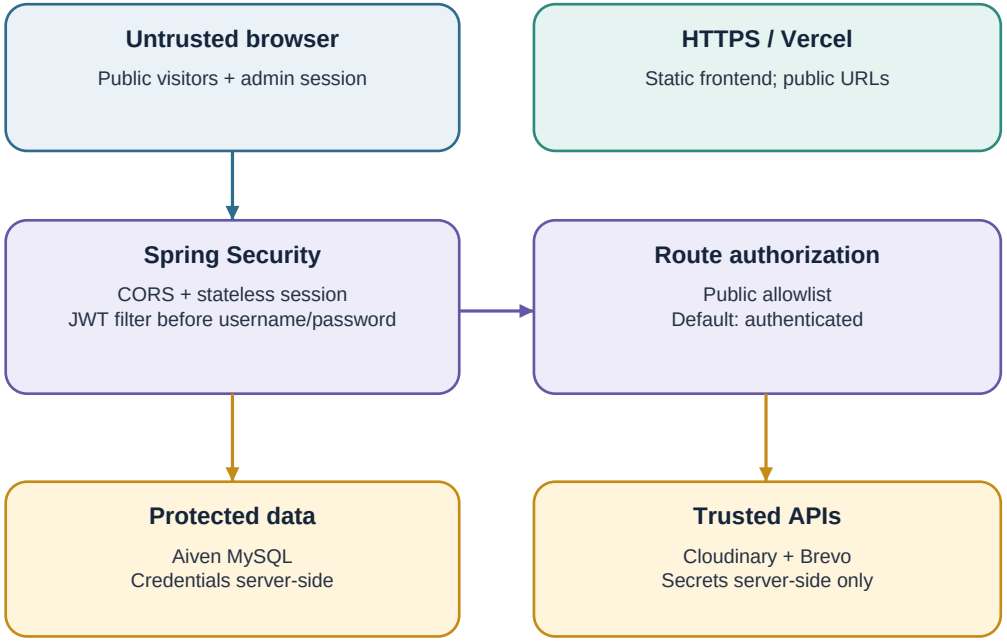
PERSISTENCE WARNING

Render Free web-service filesystems are ephemeral: local uploads can disappear after spin-down/restart/redeploy. Cloudinary was introduced to avoid this for new/replaced uploads, but migration of 44 older assets was intentionally skipped. Audit existing DB URLs for old local paths before claiming all media is durable.

Security Architecture and Review

Implemented controls are not a blanket guarantee; several critical properties require source/dashboard inspection

TRUST BOUNDARIES



Control	Evidence / status	Review item
HTTPS	Provider URLs use HTTPS	Verify any custom domains / no mixed content.
Route allowlist	Public GET and selected POST rules supplied	Review new endpoints and auth rules every release.
JWT	JWT filter registered	Inspect signature, exp/issuer/audience, error handling, and roles.
Password hash	BCryptPasswordEncoder configured	Confirm stored hashes; never log credentials.
Stateless API	STATELESS configured; CSRF disabled	Appropriate only if mutation auth is token-based and not cookie-based.
CORS	CORS_ALLOWED_ORIGINS + method list	Supplied code treats environment value as one origin; do not comma-join origins. Confirm exact production origin; inspect PATCH usage.
Uploads	5 MB limit reported; Cloudinary path tested	Verify file extension/signature/type checks, authorization and rate limits.
Secrets	Render environment variables used	Never commit .env or expose Cloudinary/Brevo/JWT/DB secrets to Vite.
Database	Managed MySQL in Aiven	Verify TLS, hostname, least privilege and backup restore process.

Reliability, Monitoring and Free-Tier Boundaries

Useful mitigations reduce risk; they do not provide an uptime guarantee or auto-recover a powered-off database

Render Free web service

- Spins down after 15 min idle
- Wake-up can take about one minute
- Ephemeral filesystem
- 750 free instance hours/workspace/month

Aiven MySQL Free

- No fixed expiry, but inactivity shutdown is possible
- 1 node; 1 GB RAM + disk
- Max connections: 76
- No 99.99% SLA

UptimeRobot + backups

- HTTP(S) endpoint monitor / email alert
- Direct HEAD returned 200 after fix
- Monitor does not heal service
- Keep independent SQL/media backups

What happens during an outage

1. Request arrives
Render may be asleep or unavailable

→

2. Backend starts
Loads env; needs DB connectivity

→

3. DB lookup
If Aiven is off/unreachable, data request fails

→

4. Monitor alerts
UptimeRobot reports HTTP failure

LOW-COST OPERATING CHECKLIST

1. Keep Aiven notification email current and check warning emails.
2. Keep the UptimeRobot Render API monitor; add a separate homepage monitor if desired.
3. On failure, inspect Aiven state and first Render startup/DB log before restarting or redeploying.
4. Preserve SQL/media backups and periodically test that the dump can be restored.
5. Remember: the Axios longer GET timeout/retry helps cold starts, not a powered-off database.

Troubleshooting Playbook

Use the symptom to pick the narrowest check first; avoid speculative production changes

Symptom	Verify first	Safe first action / prevention
Homepage loads, sections blank	Open `/api/public/profile`; browser Network tab; check response status.	Separate frontend failure from API/DB failure. Hard-refresh only after endpoint recovers.
Render 502 / no-deploy	Render Events/Logs; latest deploy status; first startup exception.	Do not redeploy repeatedly; resolve the earliest startup failure.
UnknownHostException for Aiven	Aiven service state and service hostname; Render logs.	If the existing DB is off, power on the same service. Do not recreate the DB.
Flyway cannot connect/fails	Startup logs around datasource and migration name.	Confirm DB reachability and credentials; preserve backup; never manually edit history as a first fix.
UptimeRobot 403 / browser GET works	Direct HEAD request and exact monitor timestamp/region.	HEAD permission was added to public profile. Compare after Render deploy; inspect logs if mismatch persists.
Browser CORS error	Origin, OPTIONS preflight, requested method/headers, `CORS_ALLOWED_ORIGINS`.	Use exact production origin; align methods to real frontend requests; no wildcard with credentials.
Admin request 401/403	Login result, token expiry, Authorization header and server log.	Log in again; verify JWT env/config; never share token/password.
Media missing after restart	Stored URL host/path and Cloudinary Media Library.	Repoint legacy local references to Cloudinary; verify old files individually.
Contact saved, no email	Confirm DB message row, then Brevo activity/HTTP error.	Fix sender/key/quota based on error; save-before-email means message can still exist.
Uptime shows intermittent down	Compare monitor events with direct endpoint status and Render wake/redeploy log.	A cold start can cause a transient; monitoring does not validate every page or heal services.

Architecture Decisions, Glossary and References

A concise review sheet for interviews and future maintenance

Decision	Benefit	Trade-off / when to revisit
Separate React/Vite frontend and Spring Boot API	Independent deployment; clear UI/API boundary.	CORS and multi-service debugging; reconsider if operational simplicity outweighs separation.
MySQL on managed Aiven	Relational storage without running a DB VM.	Free tier inactivity/limits/no SLA; consider paid DB tier if downtime is unacceptable.
JWT for admin API	Stateless bearer-token calls.	Token lifecycle/storage/role checks; revisit if switching to cookie-based auth or stricter threat model.
Flyway migrations	Versioned schema evolution and startup consistency.	Bad migrations can stop startup; review destructive migrations and back up data.
Cloudinary for new uploads	Avoids ephemeral Render local disk for new/replaced uploads.	Third-party quotas and access modes; audit legacy URLs and access rules.
Brevo HTTPS for contact email	Avoids SMTP connectivity dependency.	Provider quotas/key/sender configuration can fail independently of DB save.
Free hosting + UptimeRobot	Minimizes monthly cost and sends alerts.	Cold start, database inactivity and no uptime guarantee; upgrade only when reliability needs justify it.

Glossary

API: browser/server contract. CORS: cross-origin browser rule. CRUD: create/read/update/delete. Flyway: versioned schema migrations. JDBC: Java database connectivity. JWT: signed authentication token. TLS: transport encryption. Ephemeral filesystem: local files not durable across restarts/deploys. Cold start: wake-up delay after inactivity.

Official references

Render Free: https://render.com/docs/free

Aiven MySQL Free: https://aiven.io/docs/products/mysql/concepts/mysql-free-tier

Vercel Git deployments: https://vercel.com/docs/git

Cloudinary Upload API: https://cloudinary.com/documentation/upload_images

UptimeRobot pricing: https://uptimerobot.com/pricing/

Spring Boot external config: https://docs.spring.io/spring-boot/3.5/reference/features/external-config.html

Flyway migrations: https://documentation.red-gate.com/flyway/flyway-concepts/migrations